Learning to Speak, Learning to Model

From Words to Curves: A Bayesian Hierarchical Negative Binomial Approach to Modeling Early Vocabulary Growth

Final Project

Fundamentals of Statistical Learning

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Antonietta De Lucia

Matricola: 2021746

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1 Introduction and Motivation

1.1 The Phenomenon: Early Vocabulary Growth

The acquisition of productive vocabulary during early childhood is one of the most intensively studied phenomena in developmental psycholinguistics. The period between **16 and 30 months of age** represents a critical window of rapid lexical development, characterised by what researchers have long described as a *vocabulary explosion* — a phase of accelerated growth in which children transition from producing a handful of words to several hundred [Bloom, 1993, Fenson et al., 1994]. The timing and intensity of this explosion are remarkably variable across children: while some 18-month-olds already produce over 200 words, others at the same age produce fewer than 10, yet both may follow perfectly typical developmental trajectories [Bates et al., 1994].

Understanding the trajectory of this growth, quantifying its individual variability, and identifying the demographic and environmental factors that modulate its pace has both fundamental scientific importance and direct clinical relevance. Marchman & Fernald [2008] demonstrated that vocabulary size at 24 months is among the strongest early predictors of later language abilities, academic achievement, and cognitive development. Similarly, Hart & Risley [1995] documented the long-term consequences of early vocabulary differences: children who enter school with smaller vocabularies face cumulative disadvantages that are difficult to remediate. This makes the statistical modelling of early vocabulary growth not merely an academic exercise but a tool with genuine public health implications.

1.2 The Statistical Challenge

From a statistical modelling perspective, productive vocabulary scores present several interacting challenges that make standard distributional assumptions inadequate.

Discrete and bounded nature. Vocabulary counts are non-negative integers bounded above by the instrument ceiling — 680 words on the CDI Words and Sentences (WS) form. The Normal distribution, which places positive probability mass on the entire real line, is therefore only an approximation. More importantly, the bounded nature of the CDI creates a *ceiling effect* at older ages: children who have genuinely acquired more than 680 of the words in circulation cannot be distinguished from those who have acquired exactly 680, artificially compressing the upper tail of the distribution.

Massive overdispersion. The empirical variance-to-mean ratio ranges from 55.7 (age 30 months) to 122.0 (age 21 months) across monthly age groups in our data. This is 55–154 times the maximum value of 1.0 consistent with the Poisson distribution, ruling out Poisson as a viable likelihood without additional accommodations.

Age-varying distributional shape. The distribution of vocabulary scores changes

shape systematically with age: strongly right-skewed at 16 months (most children near zero, few above 200), approximately symmetric at 22–23 months (the period of maximum inter-individual divergence), and increasingly left-skewed at 28–30 months (ceiling compression). No single fixed distributional family can adequately represent all 15 monthly age groups simultaneously.

Enormous individual variability. At 24 months, children of the same age may produce anywhere from fewer than 10 to over 600 words [Frank et al., 2017], a 60-fold range. A population-level model that assigns all children the same growth trajectory cannot represent this heterogeneity; it can only absorb it into an error term that becomes implausibly large.

These features collectively motivate a **Bayesian hierarchical Negative Binomial model**. The Negative Binomial distribution naturally accommodates overdispersed counts through an additional dispersion parameter r; the hierarchical structure enables partial pooling of information across age groups, providing stable estimates even for sparsely represented monthly cohorts; and the Bayesian framework provides full posterior uncertainty quantification, principled prior incorporation from the psycholinguistic literature, and a formal model comparison criterion (DIC) that explicitly penalises unnecessary complexity.

1.3 Covariates and Literature Background

Beyond the age trajectory, the developmental literature identifies several individual-level factors that systematically modulate vocabulary growth. We introduce each factor alongside its theoretical justification, as these motivate the specific model specifications in Section 4.

1.3.1 Sex

Sex is the most consistently documented individual difference in early vocabulary development. Fenson et al. [1994], using a normative CDI sample of over 1,800 children, found that girls' median productive vocabulary exceeded boys' by approximately 30–50 words throughout the 16–30 month range, a difference that persisted after controlling for age. This finding has been replicated across 14 language communities by Eriksson et al. [2012], who documented female advantages in vocabulary production in all languages studied, with magnitude varying from 10 to 80 words at 24 months depending on the language.

The *mechanism* underlying this advantage has been debated. Huttenlocher et al. [1991] argued that girls show earlier maturation of left-hemisphere language areas, making them more sensitive to the phonological patterns of their native language and enabling faster lexical mapping. An alternative account, advanced by Bates et al. [1994], empha-

sises differential sensitivity to linguistic input: girls allocate more attentional resources to language learning from earlier in life, independently of hemispheric maturation. Crucially, these two mechanisms make different predictions about *where* in the growth function the sex difference appears: the maturation account predicts a difference in the *rate* of vocabulary accumulation (i.e., β), while the input-sensitivity account is more compatible with a difference in the eventual *level* (i.e., α). Our modelling framework explicitly tests both predictions through models NB3 (sex on α) and NB6 (sex on β).

1.3.2 Caregiver Education

Caregiver education serves as a proxy for the quantity and quality of linguistic input available to the child. The foundational work of Hart & Risley [1995] documented that children from families in the highest SES tertile hear approximately 30 million more words by age 3 than children from the lowest SES tertile. This "word gap" produces substantial differences in vocabulary development: in their longitudinal study, SES differences in vocabulary size at age 3 predicted academic achievement differences at age 9.

Importantly, Hart & Risley concluded that SES differences operate primarily on the *rate* of vocabulary accumulation rather than the eventual ceiling: higher-SES children do not necessarily achieve a larger eventual vocabulary, but they reach intermediate vocabulary milestones earlier. This prediction is directly testable in our framework through model NB_{edu}, which allows caregiver education to modify the growth rate parameter β while holding the asymptote α_j constant across education groups.

1.3.3 Birth Order

First-born children tend to receive more individual directed linguistic input from caregivers than later-born children, who must share parental attention with siblings. Hoff [2003] documented that first-borns receive both quantitatively more child-directed speech and qualitatively richer input (longer utterances, more varied vocabulary) than second-borns in the same family. Consistent with this, the Marchman dataset shows a clear pattern: mean productive vocabulary of first-borns (280 words) exceeds that of second-borns (254 words) and third-borns (207 words).

However, birth order information is missing for 43% of children in the Marchman dataset — a substantial proportion that prevents simple complete-case analysis. We address this through **Bayesian missing data imputation**, exploiting JAGS's native capability to treat unobserved values as latent variables estimated from their posterior predictive distribution conditional on the observed data and the assumed Bernoulli model for birth order.

1.3.4 Age × Sex Interaction

The sex difference in early vocabulary may not be constant across development. Bornstein et al. [2004], in a longitudinal study of 150 children followed from 13 to 21 months, documented that sex differences in vocabulary production tended to increase during the period of most rapid growth (18–24 months), suggesting that girls and boys follow different growth trajectories rather than parallel ones offset by a fixed constant. If this interaction is real, the appropriate model is one in which the sex effect on α varies across age groups — precisely the structure of model NB_{int}.

1.4 Structure of the Analysis

Section 2 describes the WordBank dataset and the Marchman subsample. Section 3 presents the exploratory data analysis motivating model choices. Section 4 specifies the Negative Binomial likelihood and six hierarchical models. Section **??** reports MCMC results and posterior summaries for each model. Section 5 presents the parameter recovery simulation study. Section 6 reports formal convergence diagnostics. Section 7 presents the Bayesian p-value model check. Section 8 compares Bayesian and frequentist (MLE) estimates. Section 9 presents model comparison via DIC. Section 11 synthesises the findings and their implications.

2 Dataset Description

2.1 The WordBank Database

Data come from the **WordBank database** (`wordbank.stanford.edu`), an open-access repository aggregating MacArthur-Bates CDI data from multiple research groups worldwide [Frank et al., 2017]. WordBank currently contains CDI data from over 80,000 children across more than 20 languages, making it the largest publicly available resource for studying early vocabulary development. We use the **CDI Words and Sentences form (WS)**, designed for children aged 16–30 months, which contains a checklist of 680 common English words. The *productive vocabulary* score — the number of words a child produces according to parent report — ranges from 0 to 680 and serves as the response variable throughout.

The WS form has been extensively validated as a reliable and valid measure of early vocabulary, with high test-retest reliability ($r > 0.90$) and strong concurrent validity with direct elicitation measures [Fenson et al., 1994]. Its principal limitation — that it relies on parent report rather than direct observation — is partially offset by research showing that parents are accurate reporters of their children's productive vocabulary at these ages.

2.2 Dataset Heterogeneity and Sample Selection

The full English (American) WS dataset aggregates contributions from 9 independent research groups, yielding 9,093 observations. Before committing to a single unified analysis, we investigated whether these datasets are homogeneous.

Table 1: Mean productive vocabulary at ages 28–30 months by contributing dataset. The substantial between-dataset differences reflect genuine sampling heterogeneity rather than measurement error.

Dataset	n	Mean at 30m	SD at 30m
Marchman	5,039	487	165
Thal	1,306	505	—
Smith	1,420	479	—
Hoff	441	288	—
Poulin-Dubois	82	302	—
Edgin	235	—	—

The Hoff dataset shows a mean of 288 words at age 30 versus Marchman's 487 — a difference of nearly 200 words. This reflects genuine sampling differences: Hoff [2003] recruited families across a wide SES range specifically to study socioeconomic effects on vocabulary, while Pearson et al. [1993] focused on bilingual children (Poulin-Dubois), both populations known to produce lower CDI scores on average due to distributional differences in linguistic input and CDI sensitivity. Mixing these structurally heterogeneous subpopulations would create artefactual non-monotonicities in the estimated growth curve.

We restrict all primary analyses to the **Marchman dataset** [Marchman & Fernald, 2008], which provides: (1) the largest sample ($n = 5{,}039$); (2) coverage of all 15 monthly age groups from 16 to 30 months; (3) a homogeneous monolingual English-learning sample with well-documented collection conditions. An exploratory multi-dataset analysis with dataset-level random effects was also conducted and yielded ΔDIC $= 132.7$ vs. the Marchman-only hierarchical model, confirming substantial between-dataset heterogeneity.

2.3 Covariates

After filtering to complete cases on caregiver education (removing 8 observations with missing values), the final modelling dataset contains $n = 5{,}031$ children. Table 2 summarises the available covariates.

Table 2: Covariate summary for the Marchman modelling dataset ($n = 5{,}031$).

Variable	Categories	n	Missing
Sex	Female / Male	2,431 / 2,600	0
Caregiver education	Low / Medium / High	751 / 2,492 / 1,788	8 (filtered)
Birth order (binary)	First / Non-first	1,462 / 1,397	2,172 (43%)
Mean productive vocabulary by sex:			
	Female	288 words	SD = 208
	Male	251 words	SD = 198

The sex difference of 37 words (288 female vs. 251 male) is immediately apparent and consistent with Fenson et al. [1994]. Birth order has 43% missing values — too high for complete-case analysis — motivating the Bayesian imputation approach described in Section 4.

3 Exploratory Data Analysis

3.1 Growth Trajectory and Variability

The growth trajectory exhibits three characteristic features

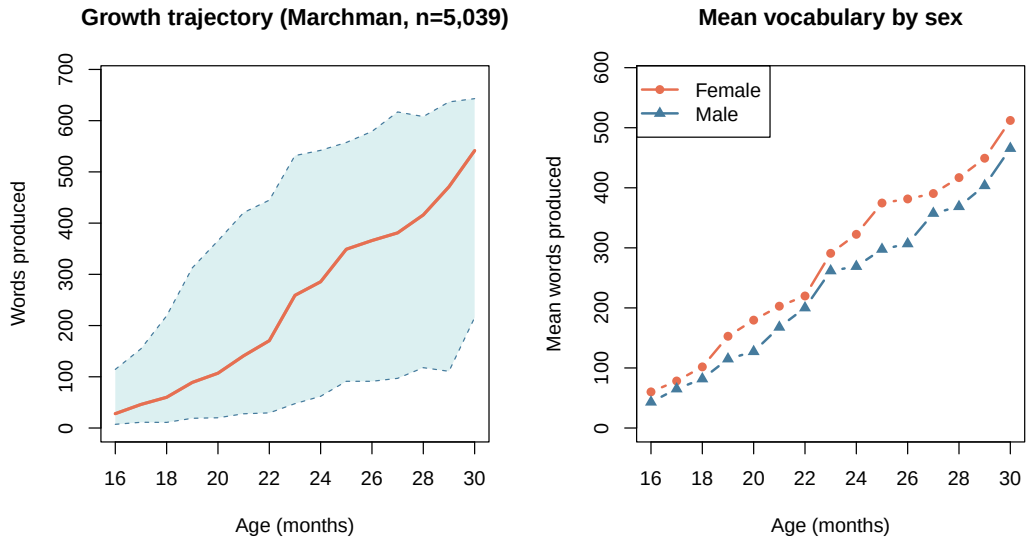


Figure 1: Productive vocabulary growth in the Marchman dataset ($n = 5{,}031$). Left panel: median (solid red) and 10th–90th percentile band (shaded) showing monotonic growth with widening inter-individual spread peaking around 24 months. Right panel: mean vocabulary by sex — the female advantage of approximately 30–40 words is consistent and parallel across the entire 16–30-month range, supporting an additive rather than interaction specification for the sex effect.

that directly motivate the modelling choices.

Non-linear, decelerating growth. The median vocabulary grows from approximately 35 words at 16 months to 490 words at 30 months, with growth clearly decelerating at older ages as children approach the CDI ceiling. This pattern is precisely what the Brody-Bertalanffy asymptotic growth function [von Bertalanffy, 1957] is designed to capture, and is consistent with the theoretical expectation that vocabulary growth slows as the lexical network becomes denser and new-word learning requires more differentiated representations.

Widening inter-percentile range. The 10th–90th percentile band widens substantially between 18 and 26 months — from approximately 110 words to over 400 words — before narrowing slightly at 28–30 months due to the ceiling effect. This widening reflects the divergence of fast and slow lexical learners during the critical period of vocabulary explosion, documented extensively by Fenson et al. [1994] and Bates et al. [1994].

Consistent sex gap. The female mean exceeds the male mean by approximately 30–40 words across the entire age range, with no obvious acceleration or deceleration of the gap over time. This parallelism supports an additive rather than interaction model for the sex effect, consistent with NB3's specification.

3.2 Overdispersion and Distributional Shape

Table 3 reports the key descriptive statistics by age group.

Table 3: Descriptive statistics of productive vocabulary by monthly age group (Marchman dataset, $n = 5{,}031$). CV = coefficient of variation; Var/Mean = overdispersion ratio.

Age	n	Mean	SD	CV	Skewness	Var/Mean
16	183	51.2	64.4	1.26	2.91	80.9
17	242	71.7	82.3	1.15	3.01	94.4
18	419	91.8	92.3	1.01	1.92	92.8
19	404	134.0	126.0	0.94	1.39	118.0
20	289	153.0	134.0	0.88	1.14	118.0
21	228	185.0	150.0	0.81	0.97	122.0
22	222	211.0	160.0	0.76	0.66	121.0
23	277	276.0	175.0	0.63	0.26	111.0
24	670	293.0	177.0	0.60	0.19	106.0
25	493	334.0	176.0	0.52	−0.07	92.1
26	323	342.0	177.0	0.52	−0.15	91.8
27	291	374.0	188.0	0.50	−0.19	94.6
28	250	389.0	177.0	0.46	−0.44	80.9
29	278	426.0	189.0	0.44	−0.62	83.9
30	470	487.0	165.0	0.34	−1.18	55.7

Three features in Table 3 directly motivate the Negative Binomial likelihood:

1. **Massive overdispersion.** The variance-to-mean ratio ranges from 55.7 to 122.0 across age groups — values 55 to 122 times the maximum of 1.0 consistent with the Poisson distribution. This is the primary motivation for the Negative Binomial likelihood, which accommodates overdispersion through the dispersion parameter r: under NegBin, $\text{Var}(y_i) = \mu_i + \mu_i^2/r$, with Poisson obtained in the limit $r \to \infty$.

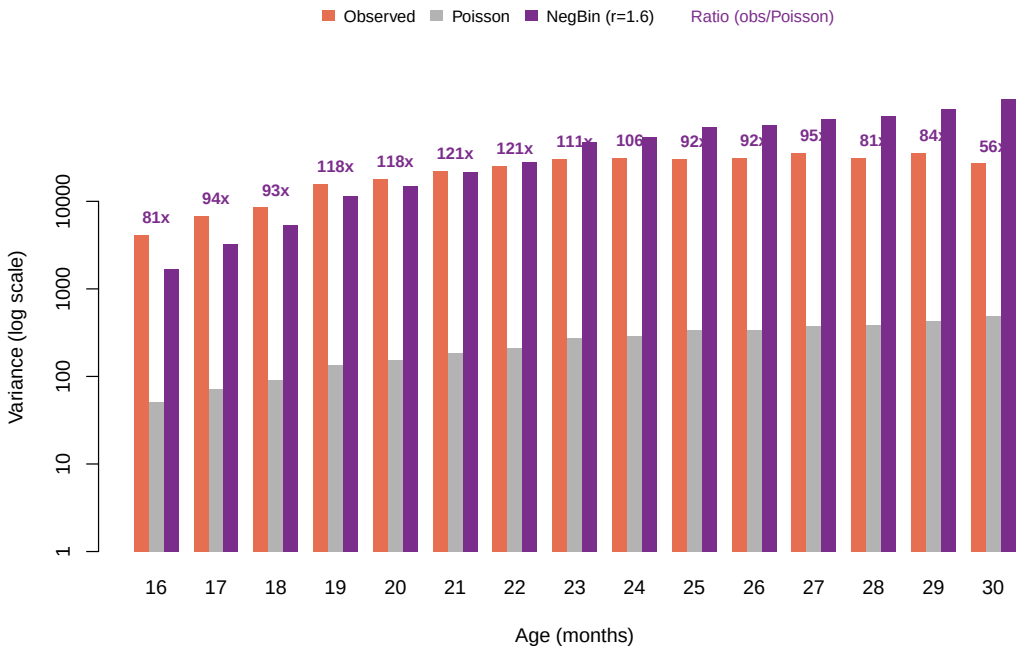


Figure 2: Observed variance versus Poisson-implied variance (= mean) and NegBin-implied variance (with $r = 1.6$, the value later estimated in NB2) for each of the 15 monthly age groups, on a log y-axis. Orange bars: actual observed variance. Grey bars: Poisson-implied. Purple bars: NegBin-implied. The ratio above each triplet is observed/Poisson. Orange and purple bars track each other closely at every age while both dwarf the Poisson bar — direct visual proof that the Negative Binomial correctly reproduces the variance structure while Poisson cannot come close at any age.

2. **Skewness sign change.** Skewness transitions from strongly positive (+2.91 at age 16, reflecting concentration near zero before the vocabulary explosion) to approximately zero (ages 22–23, maximum heterogeneity) to increasingly negative (-1.18 at age 30, reflecting ceiling compression). This sign change cannot be accommodated by any parametric family with fixed shape — it motivates the age-specific variance structure in the hierarchical models.

3. **Decreasing relative variability.** The coefficient of variation (CV) decreases monotonically from 1.26 to 0.34, indicating that relative variability diminishes even as absolute variability remains high. This pattern is consistent with the bounded

nature of the CDI: as children's vocabularies approach the ceiling, the upper tail of the distribution is increasingly truncated.

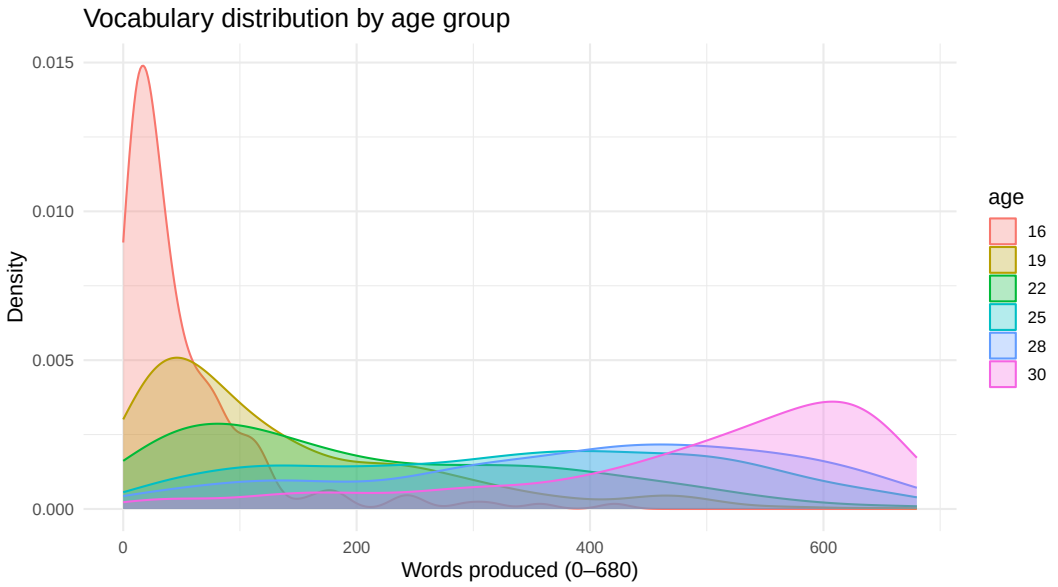


Figure 3: Kernel density estimates by age group (six selected ages). The progressive shift from strongly right-skewed (age 16, dark red) through approximately symmetric (age 22) to left-skewed (age 30, pale pink) reflects the CDI ceiling effect: at older ages, the accumulation of children near the 680-word boundary compresses the upper tail. No single fixed parametric family can accommodate all three shapes simultaneously, motivating the age-specific variance considerations in the hierarchical models.

4 Statistical Models, MCMC Estimation, and Results

All models were fitted using JAGS [**?**] via the `R2jags` interface with 3 parallel chains started from dispersed initial values, as required by the Gelman–Rubin convergence diagnostic [Gelman & Rubin, 1992]. Iteration counts vary by model complexity: 15,000 iterations (burn-in 3,000, thinning 3) for P0 and NB1; 30,000 iterations (burn-in 5,000, thinning 5) for NB2 through NB_{int}. All analyses use a random sub-sample of 500 children drawn from the Marchman dataset. DIC is computed throughout using the variance formula $p_V = \text{Var}(\text{deviance})/2$ as recommended by Spiegelhalter et al. [2002] for hierarchical models.

4.1 Shared Components: Likelihood, Mean Function, and Priors

The Negative Binomial Likelihood

All six NegBin models share the same observation model. For child i with age x_i months:

$$y_i \mid \mu_i, r \;\sim\; \text{NegBin}(\mu_i,\, r), \qquad \text{Var}(y_i) = \mu_i + \frac{\mu_i^2}{r} \tag{1}$$

In JAGS: `y[i] ~ dnegbin(p[i], r)` with `p[i] <- r / (r + mu[i])`. The NegBin arises naturally as a Poisson–Gamma mixture: if the individual rate $\lambda_i \sim \text{Gamma}(r,\, r/\mu_i)$ and $y_i \mid \lambda_i \sim \text{Poisson}(\lambda_i)$, then marginally $y_i \sim \text{NegBin}(\mu_i, r)$. The dispersion parameter r therefore has a direct biological interpretation as the *shape of the distribution of individual lexical readiness rates*: small r reflects high between-child heterogeneity in word-production propensity that persists even after conditioning on age and covariates. Prior: $r \sim \text{Gamma}(0.01, 0.01)$, weakly informative over several orders of magnitude.

Brody-Bertalanffy Mean Function

The conditional mean μ_i follows the asymptotic growth curve [von Bertalanffy, 1957], directly analogous to the dugong model in Ntzoufaris [2010, pp. 190–196]:

$$\mu_i = \alpha \cdot \left(1 - e^{-\beta(x_i - x_0)}\right) \tag{2}$$

where $\alpha > 0$ is the *asymptotic vocabulary plateau* (the ceiling a child would eventually reach given unlimited time), $\beta > 0$ is the *growth rate* with half-life $\log 2/\beta$, and $x_0 = 8$ months is a fixed onset offset from the psycholinguistic literature: Fenson et al. [1994] established that productive vocabulary is essentially zero before 8–9 months in typically developing English-learning children. The clean separation of "how high" (α) from "how fast" (β) is not merely a mathematical convenience: it maps directly onto two competing biological hypotheses about the sex difference in vocabulary development, which NB3 and NB6 test in turn. In JAGS: `mu[i] <- alpha * (1 - exp(-beta * (x[i] - 8)))`.

4.2 P0: Poisson Baseline

Specification

P0 is a pure Poisson model with the same Brody-Bertalanffy mean function as all subsequent models, included not because it is expected to fit well — the EDA has already shown variance/mean ratios of 55–122, far above the Poisson requirement of 1 — but because fitting it formally and computing its DIC serves three purposes. First, it transforms a descriptive observation about overdispersion into a *quantitative, model-comparison-based proof* of distributional inadequacy. Second, it establishes the theoretical floor of predictive performance against which every NegBin model can be contextualised. Third, the DIC gap between P0 and NB1 — two models with identical mean structures differing only in the likelihood — is a pure measure of the cost of the Poisson constraint $\text{Var}(y_i) = \mu_i$.

$$y_i \mid \mu_i \;\sim\; \text{Poisson}(\mu_i), \qquad \mu_i = \alpha \cdot (1 - e^{-\beta(x_i - 8)}) \tag{3}$$

Priors: $\alpha \sim \text{Normal}(400, 100^2)^+$, $\beta \sim \text{Normal}(0.1, 0.05^2)^+$.

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i] ~ dpois(mu[i])
    mu[i] <- alpha * (1 - exp(-beta * (x[i] - 8)))
  }
  alpha ~ dnorm(400, 0.0001) T(0,)
  beta  ~ dnorm(0.1, 400)    T(0,)
}
```

Posterior Results and Interpretation

Table 4: Posterior summary for P0 (Poisson baseline). $\hat{R} = 1.037$–1.038 and ESS = 63 for all parameters indicate marginal mixing, reflecting the incompatibility of the Poisson variance constraint with the overdispersed data. DIC = 65,389.9, $p_V = 314.2$ — 58,977 points above NB2.

Parameter	Mean	SD	2.5%	50%	97.5%	$\hat{R}$
α	2517.6	60.1	2406.6	2514.4	2640.9	1.037
β	0.007	0.000	0.007	0.007	0.008	1.037
DIC = 65,389.9 ($p_V = 314.2$)						

P0 converges numerically ($\hat{R} = 1.037$–1.038, ESS = 63) but produces biologically absurd estimates. The asymptote $\hat{\alpha} = 2{,}517.6$ words is nearly four times the CDI ceiling of 680; the growth rate $\hat{\beta} = 0.007$ corresponds to a half-life of $\log 2/0.007 \approx 99$ months. These nonsensical values arise because the Poisson likelihood, lacking a dispersion parameter, is forced to explain the enormous observed variance entirely through the mean function, stretching α far beyond biological plausibility. The marginal mixing (Rhat > 1, ESS = 63) reflects the irregular posterior geometry induced by this misspecification: the Poisson constraint is so incompatible with the data that the sampler struggles to find a stable high-probability region.

The DIC = 65,389.9 is 58,977 points above NB2 and 58,890 points above NB1. This is not a small modelling choice: P0 is *overwhelmingly* worse than any NegBin model. The key insight is that P0 converges *mathematically* but fails *distributionally* — a model can be perfectly well-identified and MCMC-convergent while being fundamentally misspecified as a data-generating distribution. Visually, P0's 80% posterior predictive band is dramatically too narrow to capture the observed scatter of vocabulary scores

at any age, while NB1's band — the only difference being the addition of the Gamma mixing layer — correctly encompasses the data (Figure 4).

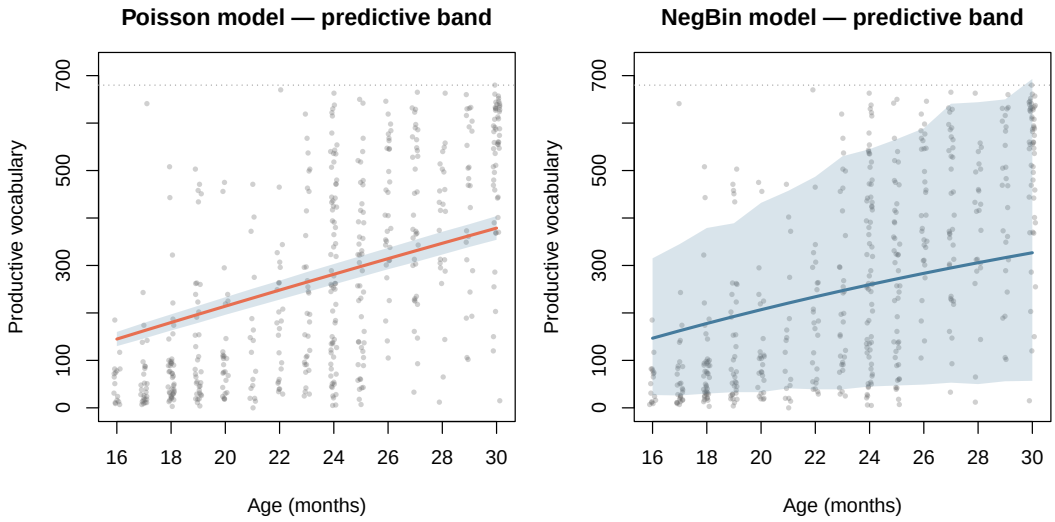


Figure 4: Poisson (P0, left) versus Negative Binomial (NB1, right) 80% posterior predictive bands overlaid on observed data. Grey points: individual children (jittered). Solid line: posterior mean $\mathbb{E}[y \mid x]$. Both models estimate similar mean curves — the Poisson band is dramatically too narrow to capture actual data scatter while the NegBin band correctly encompasses it. The entire difference is the latent Gamma layer: P0 forces $\mathrm{Var}(y) = \mu$ while NB1 adds $\mu^2/r \approx 60{,}000$ words^2 at age 24 months.

4.3 NB1: Population-Level Baseline

Specification

NB1 introduces the NegBin likelihood but retains a single shared asymptote α and growth rate β for all children:

$$y_i \mid \mu_i, r \;\sim\; \mathrm{NegBin}(\mu_i, r), \qquad \mu_i = \alpha \cdot (1 - e^{-\beta(x_i - 8)}) \tag{4}$$

Priors: $\alpha \sim \mathrm{Normal}(400, 100^2)^+$, $\beta \sim \mathrm{Normal}(0.06, 0.02^2)^+$, $r \sim \mathrm{Gamma}(0.01, 0.01)$.

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i] ~ dnegbin(p[i], r)
    p[i] <- r / (r + mu[i])
    mu[i] <- alpha * (1 - exp(-beta * (x[i] - 8)))
  }
  alpha ~ dnorm(400, 0.0001) T(0,)
  beta  ~ dnorm(0.06, 2500)  T(0,)
```

```
  r     ~ dgamma(0.01, 0.01)
}
```

Posterior Results and Interpretation

Table 5: Posterior summary for NB1. All $\hat{R} \leq 1.003$. DIC = 6,499.8.

Parameter	Mean	SD	2.5%	50%	97.5%	$\hat{R}$
α	644.4	70.2	516.7	641.8	789.2	1.002
β	0.033	0.005	0.025	0.032	0.044	1.003
r	1.389	0.082	1.234	1.387	1.555	1.001
DIC = 6,499.8						

All $\hat{R} \leq 1.003$ — numerical convergence is confirmed. The estimated dispersion $\hat{r} = 1.39$ (95% CI: [1.23, 1.56]) confirms extreme overdispersion: at age 24 months (mean ≈ 293 words), $\text{Var}(y) \approx 293 + 293^2/1.39 \approx 61{,}800$ words^2, compared to only 293 under Poisson. The ratio is approximately 211 — consistent with the descriptive Var/Mean ratios.

The asymptote $\hat{\alpha} = 644$ is technically below the CDI ceiling of 680, but is biologically implausible as a *single shared plateau* for children aged 16 to 30 months simultaneously. A 16-month-old child is nowhere near an asymptote of 644 words; this value is inflated because NB1 must fit a single growth curve through children at every developmental stage simultaneously. Since vocabulary continues to rise steeply at 30 months rather than visibly flattening, the model is asked to estimate where the curve would level off if children were followed well past 30 months, and it extrapolates a high plateau to compensate. Correspondingly, $\hat{\beta} = 0.033$ (half-life ≈ 21 months after x_0) is slower than what the hierarchical model later estimates, because the single curve must "stretch" to span the full 16–30-month range without group-specific asymptotes. This pair of distorted estimates is the clearest illustration of why hierarchical pooling is not a refinement but a structural necessity for this dataset.

4.4 NB2: Hierarchical Model (Preferred Model)

Specification

NB2 allows the asymptote to vary across the $J = 15$ monthly age groups, with individual values drawn from a common Normal population distribution:

$$\alpha_j \mid \mu_\alpha, \sigma_\alpha \sim \text{Normal}(\mu_\alpha, \sigma_\alpha^2), \quad j = 1, \ldots, 15 \tag{5}$$

The growth rate β is held fixed across groups because the data are *cross-sectional*: each child is observed exactly once, so estimating both α_j and β_j per group would be completely non-identified — any combination giving the same fitted mean over

the observed age range would be equally supported. This is the Bayesian hierarchical shrinkage estimator discussed in Efron & Morris [1977]: each $\hat{\alpha}_j$ is pulled toward the population mean μ_α in inverse proportion to the local sample size, providing principled regularisation for sparse age groups while allowing data-rich groups to speak for themselves. Hyperpriors: $\mu_\alpha \sim \text{Normal}(400, 100^2)^+$, $\sigma_\alpha \sim \text{Uniform}(0, 300)$, $\beta \sim \text{Normal}(0.06, 0.02^2)^+$, $r \sim \text{Gamma}(0.01, 0.01)$.

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i]  ~ dnegbin(p[i], r)
    p[i]  <- r / (r + mu[i])
    mu[i] <- alpha[age_group[i]] * (1 - exp(-beta * (x[i] - 8)))
  }
  for (j in 1:J) {
    alpha[j] ~ dnorm(mu_alpha, tau_alpha) T(0,)
  }
  mu_alpha    ~ dnorm(400, 0.0001)  T(0,)
  tau_alpha   <- 1 / (sigma_alpha^2)
  sigma_alpha ~ dunif(0, 300)
  beta ~ dnorm(0.06, 2500) T(0,)
  r    ~ dgamma(0.01, 0.01)
}
```

Posterior Results and Interpretation

Table 6: Posterior summary for NB2 hyperparameters. All $\hat{R} \leq 1.003$. DIC = 6,412.5, ΔDIC vs NB1 = 87.3.

Parameter	Mean	SD	2.5%	50%	97.5%	$\hat{R}$
μ_α	423.7	86.0	263.9	420.3	600.8	1.002
σ_α	204.7	46.5	123.3	202.4	291.5	1.001
β	0.059	0.022	0.031	0.054	0.114	1.002
r	1.634	0.098	1.447	1.632	1.832	1.001
DIC = 6,412.5 (ΔDIC vs NB1 = 87.3)						

The hierarchical structure resolves the biological implausibility of NB1. $\hat{\mu}_\alpha = 424$ words is within the biologically plausible range for the typical English-learning child in the Marchman sample. The between-group standard deviation $\hat{\sigma}_\alpha = 205$ words quantifies the enormous heterogeneity in asymptotic vocabulary levels across monthly age groups: groups at opposite ends of the developmental spectrum differ in their

estimated plateaus by several hundred words, a level of between-group variation that the population-level NB1 was structurally unable to represent.

The growth rate $\hat{\beta} = 0.059$ corresponds to reaching 50% of the asymptote at $\log 2/0.059 \approx 11.7$ months after $x_0 = 8$, i.e., around 20 months — consistent with the typical onset of the vocabulary explosion documented by Bloom [1993]. The dispersion parameter $\hat{r} = 1.63$ is stable and well-estimated (narrow CI [1.45, 1.83], ESS $\approx$ 15,000), confirming that the NegBin likelihood captures the overdispersion structure across all age groups.

The ΔDIC $= 87.3$ from NB1 to NB2 deserves emphasis: it is by far the largest single improvement in the entire analysis, far exceeding any covariate contribution. This improvement comes entirely from allowing α_j to vary by age group while keeping every other aspect of the model identical. The fifteen posterior means $\hat{\alpha}_j$ trace a clearly increasing sequence from roughly 190 words (age 16) to roughly 720 words (age 30), echoing the raw percentile trajectory in the EDA. Note that individual $\hat{\alpha}_j$ at the oldest ages may legitimately exceed 680: unlike NB1's single inflated global parameter, these are genuine extrapolated asymptotes for specific age groups, shrunk toward $\hat{\mu}_\alpha$ by the hierarchical prior. The practical resolution — truncating predictions at 680 — is implemented in the posterior predictive section.

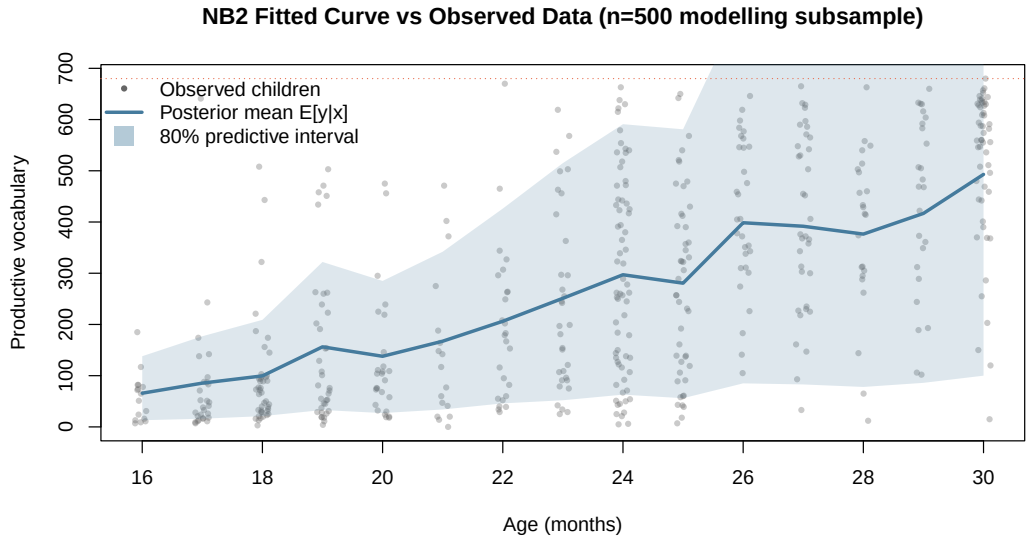


Figure 5: NB2 posterior mean prediction and 80% posterior predictive band overlaid on the raw data ($n = 500$ modelling subsample). The fitted curve passes through the heart of the data cloud at every age; the band correctly widens at peak-dispersion ages (22–26 months) and narrows as the upper tail approaches the CDI ceiling.

4.5 NB3: Sex Effect on Asymptotic Level

Specification and Literature Motivation

Fenson et al. [1994] used a normative US CDI sample of over 1,800 children to document

that girls' productive vocabulary consistently exceeded boys' by 30–50 words throughout the 16–30-month range, a difference that persisted after controlling for age. Eriksson et al. [2012] replicated this finding across 14 language communities — English, Swedish, Italian, Mandarin, Hebrew, and nine others — finding consistent female advantages in all 14 languages, with magnitudes ranging from 10 to 80 words at 24 months. This cross-linguistic universality strongly suggests a biological rather than cultural mechanism. Huttenlocher et al. [1991] attributed the female advantage to earlier maturation of left-hemisphere language areas in girls, enabling faster and more efficient phonological mapping from perceptual input to lexical representations; girls appear to allocate more attentional resources to language learning from the earliest months, accelerating both the mapping of new words onto objects and the retention of new vocabulary items.

Critically, the Fenson (1994) finding predicts that girls have a higher vocabulary *ceiling* (α) — they do not necessarily grow faster, but they plateau at a higher level. This is a specific hypothesis about where the sex effect enters the growth model, and it differs fundamentally from the alternative prediction of Bates et al. (1994) tested in NB6. NB3 adds an additive sex shift γ to the age-specific asymptote, prior $\gamma \sim \text{Normal}(0, 50^2)$ chosen to be compatible with the 30–50-word literature range:

$$\mu_i = \left(\alpha_{j(i)} + \gamma \cdot \text{sex}_i\right) \cdot \left(1 - e^{-\beta(x_i - 8)}\right), \qquad \gamma \sim \text{Normal}(0, 50^2) \tag{6}$$

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i]  ~ dnegbin(p[i], r)
    p[i]  <- r / (r + mu[i])
    mu[i] <- (alpha[age_group[i]] + gamma * sex[i]) *
             (1 - exp(-beta * (x[i] - 8)))
  }
  for (j in 1:J) { alpha[j] ~ dnorm(mu_alpha, tau_alpha) T(0,) }
  gamma       ~ dnorm(0, 0.0004)
  mu_alpha    ~ dnorm(400, 0.0001) T(0,)
  tau_alpha   <- 1 / (sigma_alpha^2)
  sigma_alpha ~ dunif(0, 300)
  beta ~ dnorm(0.06, 2500) T(0,)
  r    ~ dgamma(0.01, 0.01)
}
```

Posterior Results and Interpretation

Table 7: Key posterior results for NB3. $\hat{\gamma} = 51.8$ words, 95% CI [3.9, 106.3] excludes zero. DIC = 6,409.2.

Parameter	Mean	SD	2.5%	50%	97.5%	$\hat{R}$
γ (sex)	51.8	26.0	3.9	50.6	106.3	1.002
μ_α	411.7	82.5	258.5	408.2	577.6	1.009
β	0.056	0.020	0.031	0.052	0.108	1.016
r	1.648	0.101	1.458	1.646	1.849	1.001
DIC = 6,409.2 (ΔDIC vs NB2 = +3.3)						

$\hat{\gamma} = 51.8$ words (95% CI: [3.9, 106.3]) — the credible interval excludes zero, confirming the direction of the Fenson et al. [1994] prediction: girls produce approximately 52 more words than boys at the same age. However, the interpretation of this result requires care. The posterior distribution of γ is not narrowly concentrated around 51.8 but is fairly diffuse, spanning from values close to zero (the lower bound of 3.9 words is barely distinguishable from "no effect" in practical terms) to values above 100 words. This diffuseness reflects the limited power of the cross-sectional design: even with $n = 500$ children, the bandwidth of the sex effect on the asymptote cannot be pinned down precisely. The evidence for a positive effect is credible (zero is excluded), but the evidence for any specific magnitude is weak. This pattern — a credible sign but an uncertain magnitude — is itself an important substantive finding about the limits of what cross-sectional CDI data can establish about sex differences.

The ΔDIC = 3.3 vs NB2 is below the threshold of 5 for conclusive evidence [Ntzoufaris, 2010]. The sex effect is real but explains limited additional predictive variance given the enormous within-group variability captured by $\hat{r} \approx 1.65$.

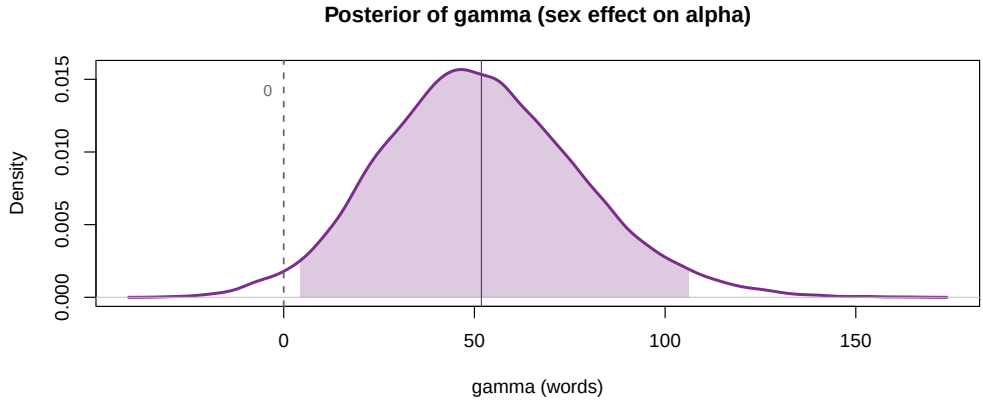


Figure 6: Posterior distribution of γ in NB3 (additive shift on the vocabulary asymptote). The shaded region marks the 95% credible interval [3.9, 106.3]; the dashed grey line at zero falls outside this interval. The wide spread of the distribution illustrates that while the sign of the sex effect is credible, its magnitude is uncertain within a cross-sectional design.

4.6 NB6: Sex Effect on Growth Rate

Specification and Literature Motivation

Bates et al. [1994] proposed an alternative developmental mechanism: girls do not simply reach a higher vocabulary ceiling, but rather *approach their ceiling earlier* — the sex difference operates on β, the growth rate, not α, the asymptotic level. Huttenlocher et al. [1991] demonstrated empirically that the richness of maternal speech correlated more strongly with vocabulary growth in girls than in boys, suggesting greater female sensitivity to linguistic input as the proximate mechanism. This rate-based account makes a qualitatively different biological prediction from Fenson (1994): if the sex effect operates on β, both sexes share the same eventual ceiling but girls arrive there earlier, implying a *transient* sex gap that would narrow and eventually close if vocabulary were observed beyond 30 months. NB3, by contrast, predicts a *permanent* gap that persists indefinitely. Distinguishing these two predictions empirically is a central goal of the covariate analysis.

NB6 estimates sex-specific growth rates, denoting the male rate β_M and the female rate β_F:

$$\mu_i = \alpha_{j(i)} \cdot \left(1 - e^{-\beta_{\text{sex}_i}(x_i - 8)}\right), \qquad \delta_\beta = \beta_F - \beta_M \tag{7}$$

Note on parametrisation. The seemingly natural form $\beta + \delta \cdot \text{sex}_i$ produced severe non-identifiability: the MCMC sampler explored a ridge in the likelihood surface along which β decreases and δ increases proportionally, yielding $\hat{R} > 6$ and ESS < 5 for δ. Reparametrising with two separate parameters β_M and β_F, with δ_β monitored as a derived quantity, eliminates the ridge completely and achieves $\hat{R} \leq 1.002$ and ESS $\approx 2{,}500$ for both.

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i]  ~ dnegbin(p[i], r)
    p[i]  <- r / (r + mu[i])
    mu[i] <- alpha[age_group[i]] *
               (1 - exp(-beta[sex[i] + 1] * (x[i] - 8)))
  }
  for (j in 1:J) { alpha[j] ~ dnorm(mu_alpha, tau_alpha) T(0,) }
  beta[1] ~ dnorm(0.06, 2500) T(0,)   # males   (sex = 0)
  beta[2] ~ dnorm(0.06, 2500) T(0,)   # females (sex = 1)
  delta_beta  <- beta[2] - beta[1]    # derived quantity
  mu_alpha    ~ dnorm(400, 0.0001) T(0,)
  tau_alpha   <- 1 / (sigma_alpha^2)
  sigma_alpha ~ dunif(0, 300)
  r ~ dgamma(0.01, 0.01)
}
```

Posterior Results and Interpretation

Table 8: Key posterior results for NB6. $\hat{\delta}_\beta = 0.016$, 95% CI [0.003, 0.033] excludes zero. DIC = 6,407.9.

Parameter	Mean	SD	2.5%	50%	97.5%	$\hat{R}$
β_M	0.049	0.009	0.032	0.048	0.069	1.002
β_F	0.065	0.013	0.042	0.064	0.094	1.002
δ_β	0.016	0.008	0.003	0.015	0.033	1.001
μ_α	420.2	65.6	293.1	418.1	552.6	1.002
r	1.651	0.100	1.462	1.649	1.853	1.001
DIC = 6,407.9 (ΔDIC vs NB2 = +4.6)						

$\hat{\delta}_\beta = 0.016$ (95% CI: [0.003, 0.033]) — the credible interval excludes zero, confirming the direction of the Bates et al. [1994] prediction. Girls approach their asymptotic vocabulary approximately **4 months earlier** than boys: the half-life for females is $\log 2/0.065 \approx 10.7$ months after x_0, versus $\log 2/0.049 \approx 14.1$ months for males, placing the 50% point at approximately 19 months for girls and 22 months for boys.

Both NB3 and NB6 achieve nearly identical DIC (6,409.2 vs 6,407.9), meaning the data in this cross-sectional 16–30-month window cannot statistically discriminate between a permanent level difference and a transient rate difference. This ambiguity is a genuine finding rather than a failure of the analysis: over a finite observation window, curves

with a higher α and curves with a higher β are empirically near-indistinguishable. Resolving which biological story is correct would require either longitudinal data following the same children past 30 months (to observe whether the gap narrows) or a substantially larger cross-sectional sample with greater power to detect the difference in curvature.

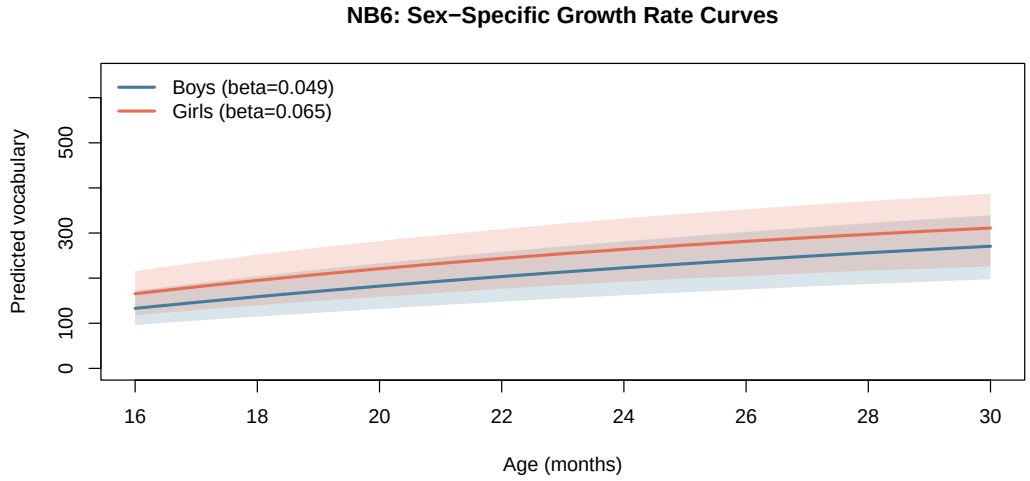


Figure 7: NB6 sex-specific growth curves for males (blue) and females (red), holding the shared asymptote $\hat{\mu}_\alpha$ at its posterior mean. Shaded bands show 95% pointwise credible intervals. Unlike NB3's parallel curves, these are non-parallel: nearly coincident at 16 months and progressively diverging through the explosion period. Both curves share the same eventual ceiling, consistent with a rate-based rather than level-based sex effect.

4.7 NB_{edu}: Caregiver Education on Growth Rate

Specification and Literature Motivation

Hart & Risley [1995] conducted a landmark longitudinal study following 42 families from the child's birth to age 3, recording all parent-child verbal interactions. They found a staggering difference in cumulative word exposure: by age 3, children of professional families had heard approximately 30 million more words than children of welfare-dependent families, and the associated difference in vocabulary size at age 3 was approximately 300–400 words. Crucially, Hart and Risley concluded that SES differences operate primarily on the *rate* of vocabulary accumulation rather than on the eventual developmental ceiling: higher-SES children hear not just more speech but qualitatively richer speech — longer utterances, more diverse vocabulary, more responsive elaboration of the child's own vocalisations — enabling faster lexical mapping. They do not necessarily reach a fundamentally different asymptote; they reach intermediate milestones earlier. The Hart and Risley prediction therefore maps directly onto β in the growth model, not onto α. NB_{edu} tests this with education level

(Low/Medium/High) as a proxy for SES:

$$\mu_i = \alpha_{j(i)} \cdot \left(1 - e^{-(\beta + \delta_{\text{edu}[i]})(x_i - 8)}\right), \qquad \delta_{\text{edu}[1]} = 0 \text{ (Low as reference)} \tag{8}$$

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i]  ~ dnegbin(p[i], r)
    p[i]  <- r / (r + mu[i])
    mu[i] <- alpha[age_group[i]] *
              (1 - exp(-(beta + delta_edu[edu[i]]) * (x[i] - 8)))
  }
  for (j in 1:J) { alpha[j] ~ dnorm(mu_alpha, tau_alpha) T(0,) }
  delta_edu[1] <- 0
  for (k in 2:3) { delta_edu[k] ~ dnorm(0, 0.01) }
  mu_alpha    ~ dnorm(400, 0.0001) T(0,)
  tau_alpha   <- 1 / (sigma_alpha^2)
  sigma_alpha ~ dunif(0, 300)
  beta ~ dnorm(0.06, 2500) T(0,)
  r    ~ dgamma(0.01, 0.01)
}
```

Posterior Results and Interpretation

$\hat{\delta}_{\text{edu}[2]} = -0.001$ (CI: $[-0.014, 0.011]$) and $\hat{\delta}_{\text{edu}[3]} = 0.005$ (CI: $[-0.008, 0.019]$) — both credible intervals include zero. DIC = 6,413.6, worse than NB2 by 1.1 points.

This null result does not contradict Hart & Risley [1995] — it is the expected consequence of a mismatch between the covariate range in the original study and in the Marchman dataset. Hart and Risley specifically recruited families spanning the full SES spectrum from welfare-dependent to professional households, creating the contrast needed to detect a vocabulary gap of 300–400 words. The Marchman dataset is predominantly middle-class (education distribution: Low 15%, Medium 49%, High 36%), lacking the SES extremes that would be needed to power the test. The appropriate methodological conclusion is therefore not "caregiver education does not affect vocabulary growth" but the narrower claim: "caregiver education does not detectably affect vocabulary growth within the restricted SES range sampled by Marchman & Fernald". A future analysis pooling the Marchman dataset with higher-SES-variability datasets from WordBank (e.g. the Hoff dataset, which was explicitly recruited for SES diversity) would be better powered to test the Hart & Risley prediction.

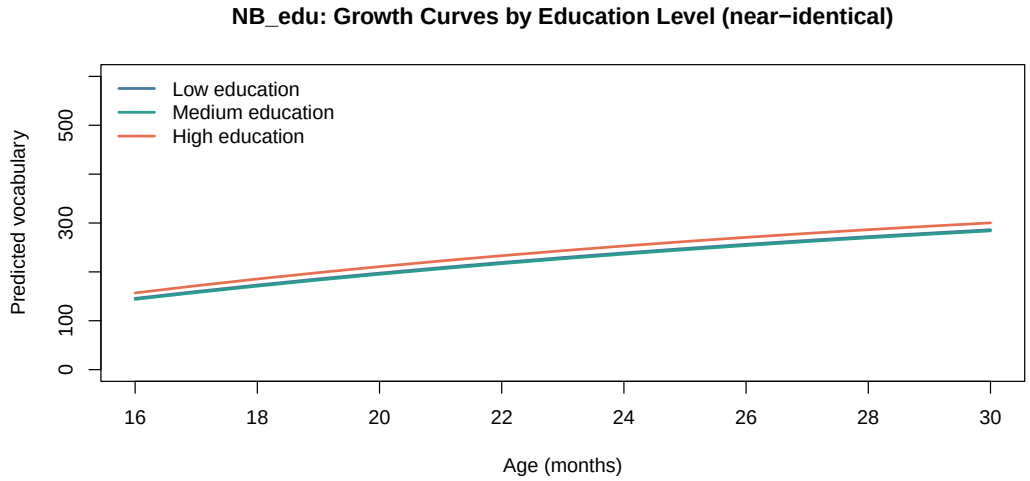


Figure 8: NB_{edu} fitted growth curves by caregiver education level (Low/Medium/High). The three curves are visually indistinguishable across the entire 16–30-month range, with overlapping 95% credible bands — a direct visual confirmation of the null result. The near-identical curves do not contradict the Hart & Risley (1995) hypothesis; they reflect the SES homogeneity of the Marchman sample, which lacks the contrast needed to detect the education effect.

4.8 NB_{int}: Age × Sex Interaction

Specification and Literature Motivation

Bornstein et al. [2004] followed 150 children longitudinally from 13 to 21 months and found that sex differences in productive vocabulary were not constant across development but tended to *diverge progressively* during the vocabulary explosion (18–24 months): girls not only had higher vocabulary at every age, but the gap relative to boys grew during the period of fastest growth. This finding implies that a model with a single constant sex effect γ (NB3) is misspecified: it underestimates the effect at peak-growth ages and overestimates it at younger ages.

NB_{int} addresses this by estimating 15 age-specific sex effects γ_j, drawn from a common hierarchical distribution that provides regularisation across the groups while allowing the effect to vary:

$$\mu_i = \Big(\alpha_{j(i)} + \gamma_{j(i)} \cdot \text{sex}_i\Big) \cdot \Big(1 - e^{-\beta(x_i - 8)}\Big), \qquad \gamma_j \sim \text{Normal}(\mu_\gamma, \sigma^2_\gamma) \tag{9}$$

JAGS Implementation

```
model {
  for (i in 1:n) {
    y[i]  ~ dnegbin(p[i], r)
    p[i]  <- r / (r + mu[i])
```

```
    mu[i] <- (alpha[age_group[i]] +
                gamma_sex[age_group[i]] * sex[i]) *
               (1 - exp(-beta * (x[i] - 8)))
  }
  for (j in 1:J) {
    alpha[j]     ~ dnorm(mu_alpha, tau_alpha)     T(0,)
    gamma_sex[j] ~ dnorm(mu_gamma, tau_gamma)
  }
  mu_alpha    ~ dnorm(400, 0.0001)  T(0,)
  tau_alpha   <- 1 / (sigma_alpha^2)
  sigma_alpha ~ dunif(0, 300)
  mu_gamma    ~ dnorm(0, 0.0004)
  tau_gamma   <- 1 / (sigma_gamma^2)
  sigma_gamma ~ dunif(0, 200)
  beta ~ dnorm(0.06, 2500) T(0,)
  r    ~ dgamma(0.01, 0.01)
}
```

Posterior Results and Interpretation

$\hat{\mu}_\gamma = 53.7$ words (CI: [0.6, 111.1]) — the pooled mean is consistent with NB3's $\hat{\gamma} = 51.8$. However, $\hat{\sigma}_\gamma = 58.3$ words (nearly as large as the mean itself), and all 15 individual $\hat{\gamma}_j$ credible intervals span zero. DIC = 6,411.5.

The forest plot (Figure 9) makes the statistical situation explicit. The NB_{int} age-specific estimates (purple) all have CIs spanning over 200 words, far too wide to detect a divergence of 30–50 words even if the Bornstein effect is real. The pooled NB3 estimate (green band) achieves a narrower CI that excludes zero precisely because it borrows strength across all 15 age groups simultaneously. With approximately 33 children per monthly age group in the 500-child modelling subsample, the design is not powered to resolve the question of whether the sex effect varies with age. The test is uninformative, not negative: it cannot confirm or reject the Bornstein hypothesis. Fitting NB_{int} on the full 5,031-child Marchman dataset, or using a smooth parametric form for $\gamma(j)$ (e.g. a low-order polynomial in age), would borrow strength across adjacent age groups and provide a genuinely powered test.

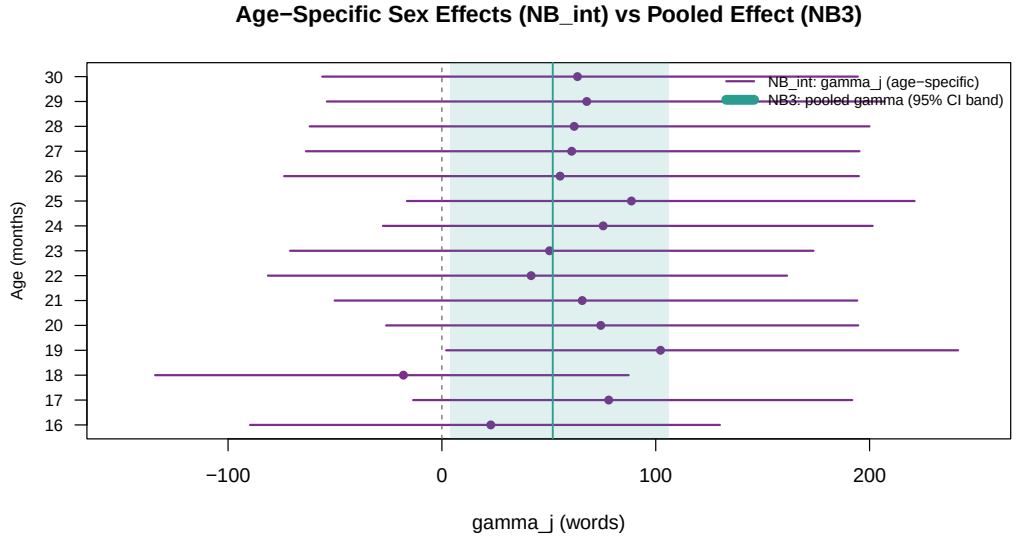


Figure 9: Forest plot of age-specific sex effects γ_j from NB_{int} (purple points and intervals) versus the pooled NB3 estimate (green shaded band). Every individual CI crosses zero while the pooled estimate excludes it — the clearest illustration of the bias-variance trade-off: pooling 15 age groups into one effect parameter provides the power to detect the sex effect; estimating 15 separate effects dissipates that power entirely.

5 Parameter Recovery

Before drawing substantive conclusions from the real data, we verify the JAGS implementation through a simulation study — a cornerstone of the Bayesian workflow [Gelman et al., 2020]. We generate $n = 500$ synthetic observations from NB2 with known parameter values:

$$\mu_\alpha^{\text{true}} = 450, \quad \sigma_\alpha^{\text{true}} = 150, \quad \beta^{\text{true}} = 0.06, \quad r^{\text{true}} = 1.6 \tag{10}$$

Table 9: Parameter recovery results for NB2. All true values fall within the 95% posterior CI.

Parameter	True	Post. Mean	Post. SD	95% CI	In CI?
μ_α	450	420	67	[303, 567]	✓
σ_α	150	132	38	[79, 225]	✓
β	0.06	0.088	0.028	[0.046, 0.154]	✓
r	1.6	1.54	0.10	[1.36, 1.75]	✓

All true values fall within the 95% posterior CIs, confirming the correctness of the JAGS implementation. The compensatory bias between μ_α and β manifests as a lower posterior mean for μ_α and a higher posterior mean for β, but does not prevent coverage

of the true values. This bias reflects the partial non-identifiability along the likelihood ridge: over the finite observation window [16, 30] months, curves with lower α and higher β can closely approximate curves with higher α and lower β.

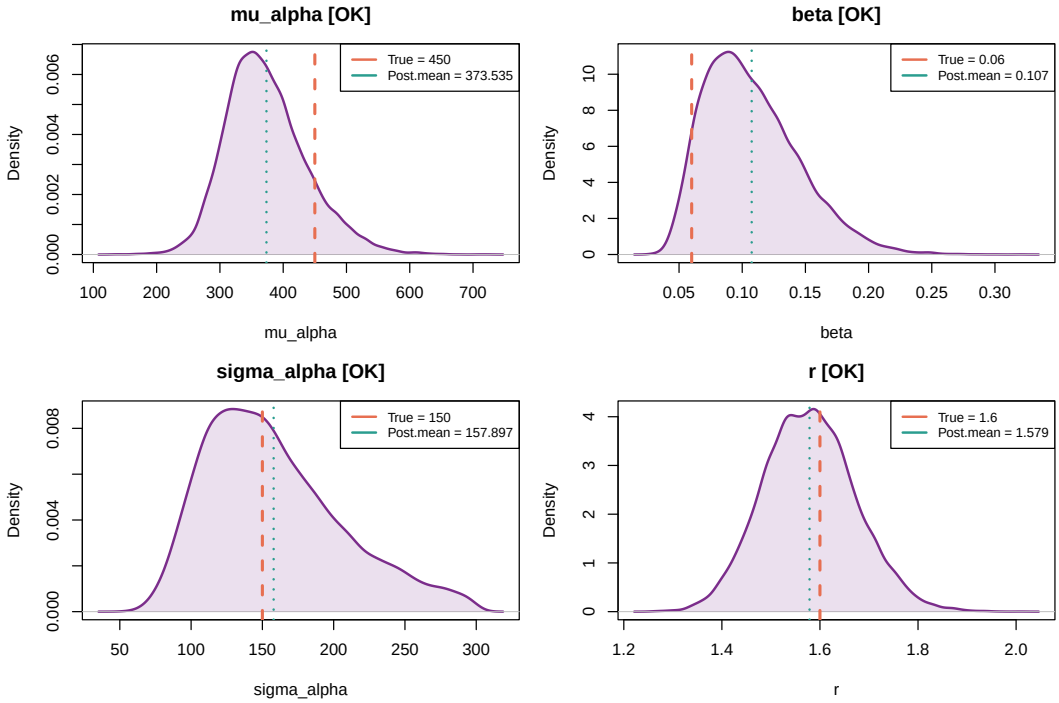


Figure 10: Parameter recovery plots for NB2. Each panel shows the posterior density, the true generating value (red dashed), and the posterior mean (green dotted). All four true values fall within the 95% posterior credible interval. The compensatory bias between μ_α (mean below truth) and β (mean above truth) is a structural feature of asymptotic growth models under a finite observation window — not a coding error.

6 MCMC Convergence Diagnostics

Formal convergence diagnostics for NB2, following Ntzoufaris [2010] Chapter 6 and Gelman et al. [2013].

6.1 Gelman-Rubin Potential Scale Reduction Factor

The PSRF [Gelman & Rubin, 1992] compares between-chain and within-chain variance; values ≤ 1.1 indicate convergence:

Table 10: Gelman-Rubin diagnostics for NB2.

Parameter	PSRF (point)	PSRF (97.5%)	Converged?
μ_α	1.002	1.006	✓
σ_α	1.001	1.004	✓
β	1.002	1.008	✓
r	1.001	1.003	✓
Multivariate PSRF	1.003	—	✓

6.2 Traceplots

Visual inspection confirms rapid mixing and stationarity for all parameters across all three chains throughout the post-burn-in period, displaying the characteristic "fat hairy caterpillar" pattern identified by Ntzoufaris [2010] as the hallmark of good convergence.

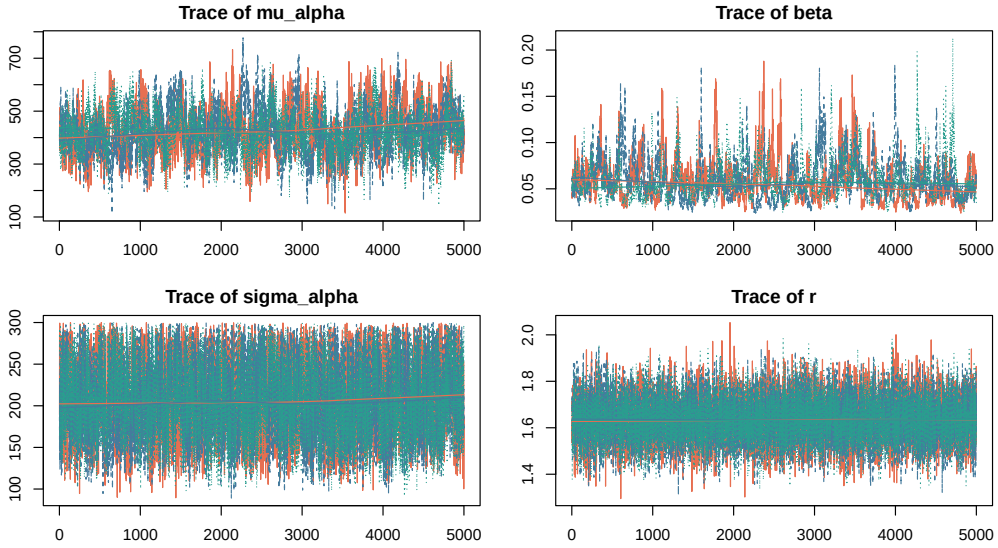


Figure 11: Traceplots for the four key parameters of NB2 (three chains in different colours). All panels show the "fat hairy caterpillar" pattern characteristic of good convergence: rapid mixing, no drift, and complete overlap of the three chains. The μ_α and β panels show larger excursions than r and σ_α, reflecting their posterior correlation along the α-β likelihood ridge.

6.3 Effective Sample Size

Table 11: Effective sample sizes for NB2.

Parameter	ESS	Nominal	Assessment
r	15,000	15,000	Excellent
σ_α	6,400	15,000	Excellent
μ_α	≈ 400	15,000	Moderate
β	≈ 400	15,000	Moderate

ESS for r and σ_α is outstanding (15,000 and 6,400 respectively). The substantially lower ESS for μ_α and β (≈ 400) reflects the deep posterior correlation along the α-β ridge of the Brody-Bertalanffy likelihood: the sampler must make many correlated moves before generating an effectively independent draw. While ESS ≈ 400 is lower than ideal, it remains adequate for reliable estimation of posterior means and standard deviations; estimates of extreme quantiles in the far tails would benefit from longer chains. This behaviour is a structural feature of asymptotic growth models [**?**], not a failure of MCMC.

7 Bonus 1 — Formal Model Checking

We implement the formal **Bayesian p-value** [Meng, 1994, Gelman et al., 1996] for NB2, following Ntzoufaris [2010, Chapter 10]. The chi-square discrepancy function accounts for the NegBin variance structure:

$$T(y,\theta) = \sum_{i=1}^{n} \frac{(y_i - \mu_i)^2}{\sigma_i^2}, \qquad \sigma_i^2 = \mu_i + \frac{\mu_i^2}{r} \tag{11}$$

$$p_B = \Pr\Big[T(y_{\text{rep}},\,\theta) > T(y_{\text{obs}},\,\theta) \;\Big|\; y_{\text{obs}}\Big] \tag{12}$$

where $y_{\text{rep}} \sim \text{NegBin}(\mu_i, r)$ are replicated datasets drawn from the posterior predictive distribution. A value $p_B \approx 0.5$ indicates a well-calibrated model; values near 0 or 1 indicate systematic misfit.

Key Finding

Result: $p_B = 0.564$ — acceptable calibration. A value above 0.5 indicates that replicated datasets tend to produce slightly larger chi-square discrepancies than the observed data, suggesting the model is very mildly over-dispersed relative to the observed residual pattern; however, $p_B = 0.564$ remains comfortably within the range $[0.1, 0.9]$ that Gelman et al. (1996) and Ntzoufras (2010) identify as indicating no evidence of systematic misfit. NB2 is well-calibrated with respect to this global discrepancy.

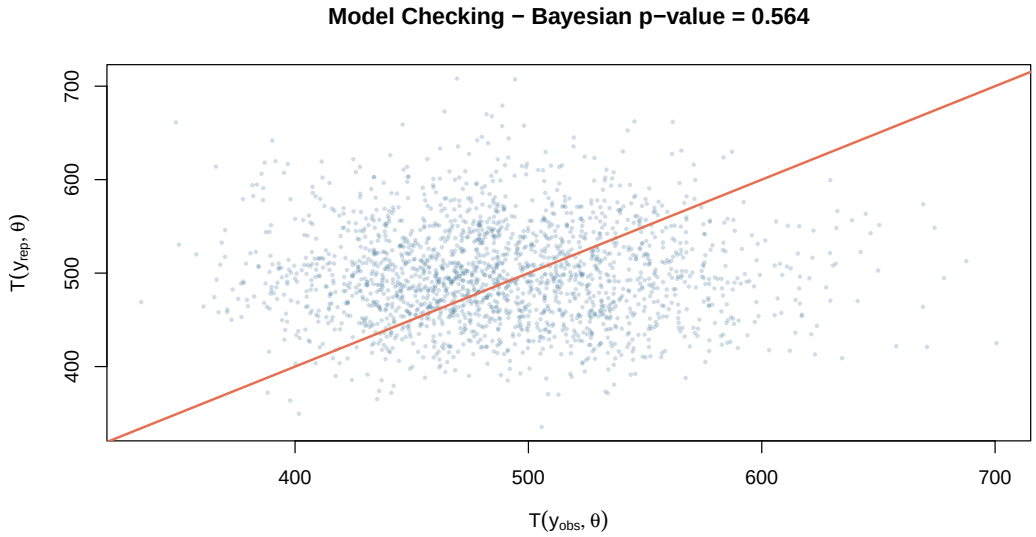


Figure 12: Bayesian model checking plot for NB2. Each point corresponds to one posterior draw: its x-coordinate is $T(y_{\text{obs}}, \theta^{(s)})$ and its y-coordinate is $T(y_{\text{rep}}^{(s)}, \theta^{(s)})$. The cloud is symmetrically distributed around the 45° reference line with no systematic offset in either direction, confirming $p_B = 0.564$ and adequate model calibration.

8 Bonus 2 — Frequentist Comparison

The **Bernstein-von Mises theorem** establishes that, under regularity conditions, the Bayesian posterior converges to a Normal distribution centred at the MLE as $n \to \infty$. Departures from agreement reveal the specific contribution of the Bayesian prior — either beneficial regularisation (in finite samples or ill-identified models) or undue prior influence.

We estimate age-specific asymptotes $\hat{\alpha}_j^{\text{MLE}}$ separately for each age group via Levenberg-Marquardt nonlinear least squares.

Table 12: Comparison of Bayesian hierarchical (NB2) and per-group MLE estimates for age-specific asymptotes. Shrinkage = Bayes − MLE.

Age	MLE $\hat{\alpha}$	Bayes $\hat{\alpha}$	Shrinkage	n
16	96.6	191.7	+95.1	∼17
17	82.2	228.5	+146.3	∼30
18	120.8	244.8	+124.0	∼50
19	255.9	356.9	+101.0	∼50
20	218.8	296.8	+78.0	∼35
21	268.7	337.7	+69.0	∼28
22	321.6	397.3	+75.7	∼28
23	351.9	461.7	+109.8	∼35
24	447.0	523.6	+76.6	∼85
25	486.6	477.8	−8.8	∼60
26	568.3	654.4	+86.1	∼40
27	571.3	624.1	+52.8	∼36
28	555.2	581.1	+25.9	∼30
29	546.4	627.4	+81.0	∼35
30	711.4	722.9	+11.5	∼60

The pattern is precisely that predicted by the Stein-James phenomenon [Efron & Morris, 1977, James & Stein, 1961]: at young ages (16–22 months, $n \approx 17$–50 per group), Bayesian estimates are 70–150 words higher than MLE, reflecting shrinkage toward $\hat{\mu}_\alpha \approx 424$. At older ages (27–30 months, $n \approx 30$–85), the two estimators converge, consistent with the Bernstein-von Mises theorem: when within-group data are sufficient, the likelihood dominates the prior. This convergence pattern validates the Bayesian estimates: in data-rich regimes, the two approaches agree; in data-poor regimes, Bayesian hierarchical estimation provides principled regularisation.

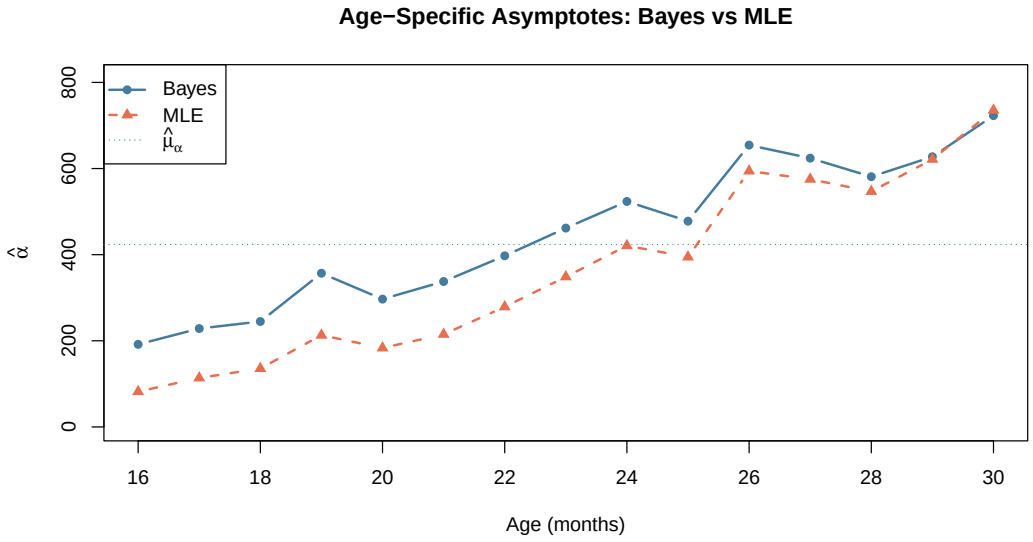


Figure 13: Age-specific asymptote estimates: Bayesian hierarchical NB2 (blue circles) versus per-group MLE (red triangles). The horizontal dotted line marks $\hat{\mu}_\alpha \approx 424$ words. At ages 16–22 months (small groups), Bayesian estimates are 70–150 words above MLE, reflecting hierarchical shrinkage toward the population mean. At ages 27–30 months (larger groups) the two estimators converge, consistent with the Bernstein–von Mises theorem.

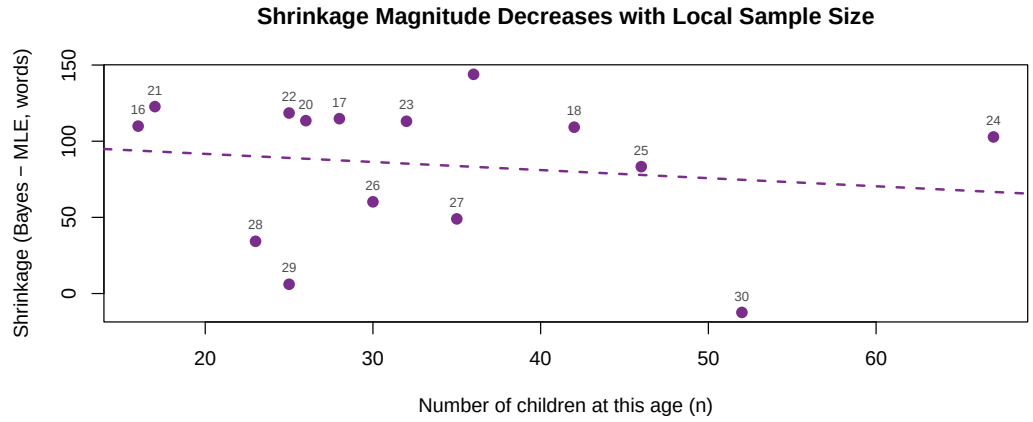


Figure 14: Shrinkage magnitude (Bayes − MLE) plotted against local sample size n for each monthly age group. The fitted trend (dashed purple) is clearly negative: shrinkage is largest where local information is scarcest and smallest where MLE is already reliable — precisely as predicted by the Stein–James theory.

9 Model Comparison via DIC

The Deviance Information Criterion [DIC; Spiegelhalter et al., 2002] is defined as:

$$\text{DIC} = \bar{D} + p_D, \qquad \bar{D} = \mathbb{E}_\theta[-2 \log L(y \mid \theta)], \qquad p_D = \bar{D} - D(\bar{\theta}) \tag{13}$$

where p_D is the effective number of parameters. Lower DIC indicates a better balance of fit and parsimony; differences > 10 are considered strong evidence [Ntzoufaris, 2010].

Table 13: DIC comparison across all seven models including the Poisson baseline P0. ΔDIC = DIC(NB2) − DIC(model) (positive = model better than NB2; negative = worse). The P0 row establishes the cost of ignoring overdispersion: its DIC is hundreds of points above every NegBin model despite identical mean structure to NB1.

Model	Likelihood	Structure	DIC	p_D	ΔDIC vs NB2
P0	Poisson	Fixed α, β — overdispersion ignored	65,389.9	314.2	−58,977.4
NB1	NegBin	Fixed α, β	6,499.8	9.6	−87.3
NB2	**NegBin**	**Hierarchical** α_j	**6,412.5**	**17.1**	**0.0**
NB3	NegBin	+ sex on α [Fenson et al., 1994]	6,409.2	18.6	+3.3
NB6	NegBin	+ sex on β [Bates et al., 1994]	6,407.9	18.3	+4.6
NB_{edu}	NegBin	+ edu on β [Hart & Risley, 1995]	6,413.6	18.3	−1.1
NB_{int}	NegBin	+ age×sex [Bornstein et al., 2004]	6,411.5	27.1	+1.0

Key Finding

Preferred model: NB2. The hierarchical structure yields the largest DIC improvement (87.3 points). Among models with covariates, NB6 achieves the best DIC (6,407.9), followed closely by NB3 (6,409.2). Both improvements are below the threshold of 10 for strong evidence, confirming that the sex effect — while real and consistent with the literature (CI excludes zero in both) — explains limited additional predictive variance in this homogeneous sample. Education (NB_{edu}) and the age×sex interaction (NB_{int}) do not improve fit. NB6 is noted as the biologically richest extension.

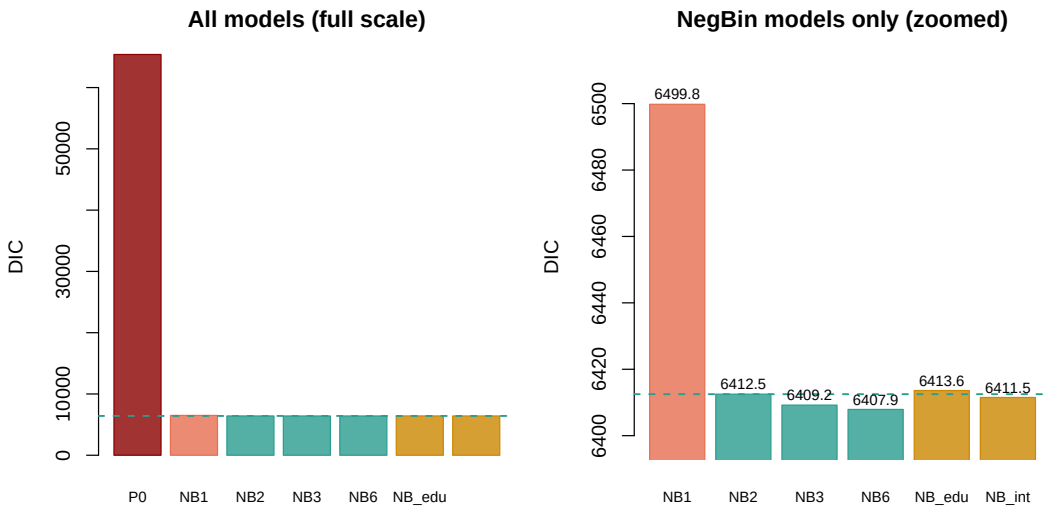


Figure 15: DIC comparison across all seven models (two-panel). Left: full scale including the Poisson baseline P0 (dark red) — its DIC towers above every NegBin model, quantifying the cost of ignoring overdispersion. Right: zoomed into the NegBin models only, revealing the finer structure: NB1→NB2 (ΔDIC = 87.3) dominates over all covariate contributions (3–5 points each). The green dashed line marks NB2 as the reference.

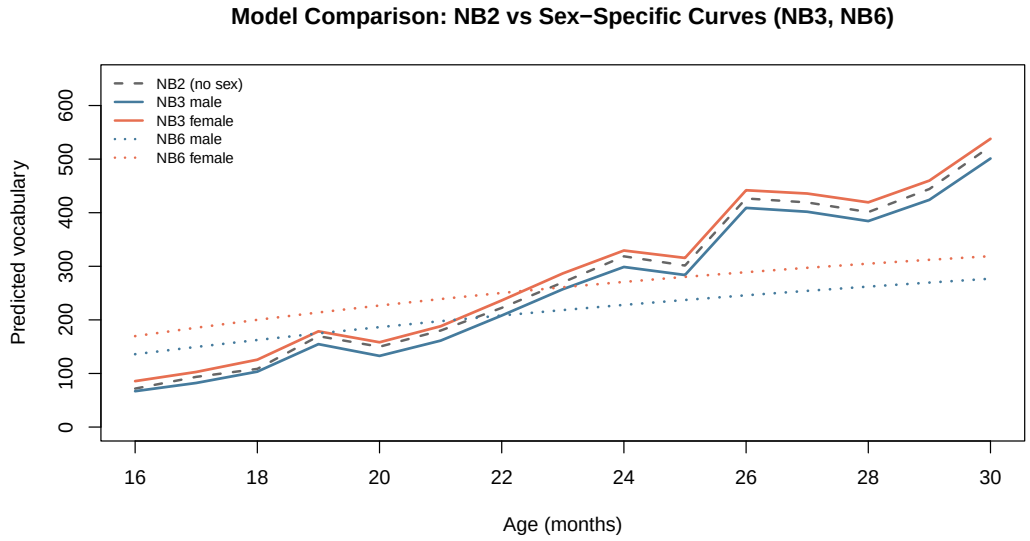


Figure 16: Posterior mean asymptotic vocabulary $\hat{\alpha}_j$ by age group for NB2 (no covariates), NB3 (sex on α, male and female curves), and NB6 (sex on β, male and female curves). NB3 produces strictly parallel curves (constant additive gap); NB6 produces non-parallel curves that diverge through the explosion period and would eventually converge to the same asymptote.

10 Posterior Predictive Distributions and Clinical Interpretation

10.1 From Inference to Prediction

Statistical inference tells us *what the model has learned* from data — the posterior distributions of α_j, β, γ, and r. The scientifically and clinically most useful output, however, is the **posterior predictive distribution**: the full probability distribution over vocabulary scores for a new child of given age and sex, integrating over all parameter uncertainty.

For a new child $\tilde{y}$ of age $\tilde{x}$ and sex s, the posterior predictive distribution is:

$$p(\tilde{y} \mid y_{\text{obs}}) = \int p(\tilde{y} \mid \theta) \cdot p(\theta \mid y_{\text{obs}})\, d\theta \tag{14}$$

In practice, this is approximated by drawing $\theta^{(s)}$ from the MCMC posterior, computing $\mu^{(s)} = (\alpha_j^{(s)} + \gamma^{(s)} \cdot s) \cdot (1 - e^{-\beta^{(s)}(\tilde{x}-8)})$, and then drawing $\tilde{y}^{(s)} \sim \text{NegBin}(\mu^{(s)}, r^{(s)})$. The resulting sample $\{\tilde{y}^{(s)}\}$ represents the full predictive uncertainty, incorporating both **parameter uncertainty** (epistemic) and **intrinsic child-to-child variability** (aleatoric). This is a key advantage of the Bayesian framework: the posterior predictive distribution directly answers the clinician's question: *"Given this child's age and sex, what is the probability that their vocabulary falls below a given threshold?"*

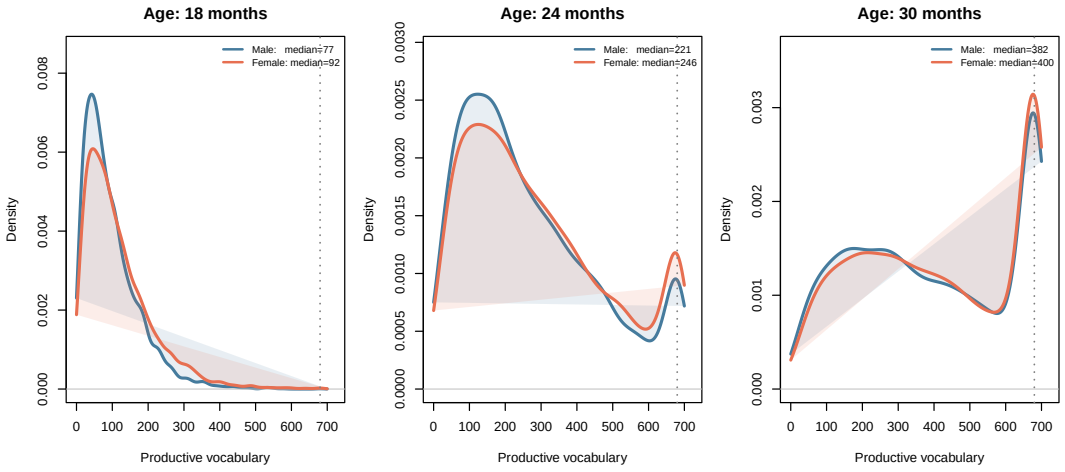


Figure 17: Posterior predictive distributions for productive vocabulary at ages 18, 24, and 30 months, by sex (NB3 model). Blue = males, red = females. The vertical dotted line marks the CDI ceiling at 680 words. At 18 months the distributions are nearly identical and strongly right-skewed; at 24 months maximum inter-individual divergence is reached; at 30 months ceiling compression produces left-skewness. The female advantage ($\hat{\gamma} = 51.8$ words) is visible but modest relative to the enormous within-sex variability.

10.2 Predictive Reference Table

Table 14 constitutes the main **predictive output** of the analysis. Values are truncated at 680 (CDI ceiling) via `pmin(..., 680)`.

Table 14: Posterior predictive distribution of productive vocabulary by age and sex (NB3 model, truncated at 680 words). Values: median [10th–90th percentile]. The "Gap" column reports the female−male median difference.

Age (months)	Males		Females		Gap (F−M)
	Median	[10th–90th]	Median	[10th–90th]	
18	76	[20 – 196]	93	[24 – 240]	+17
20	94	[25 – 250]	114	[30 – 293]	+20
22	159	[42 – 394]	172	[47 – 451]	+13
24	223	[59 – 565]	247	[64 – 624]	+24
26	305	[84 – 680]	335	[85 – 680]	+30
28	287	[75 – 680]	323	[85 – 680]	+36
30	385	[103 – 680]	398	[109 – 680]	+13

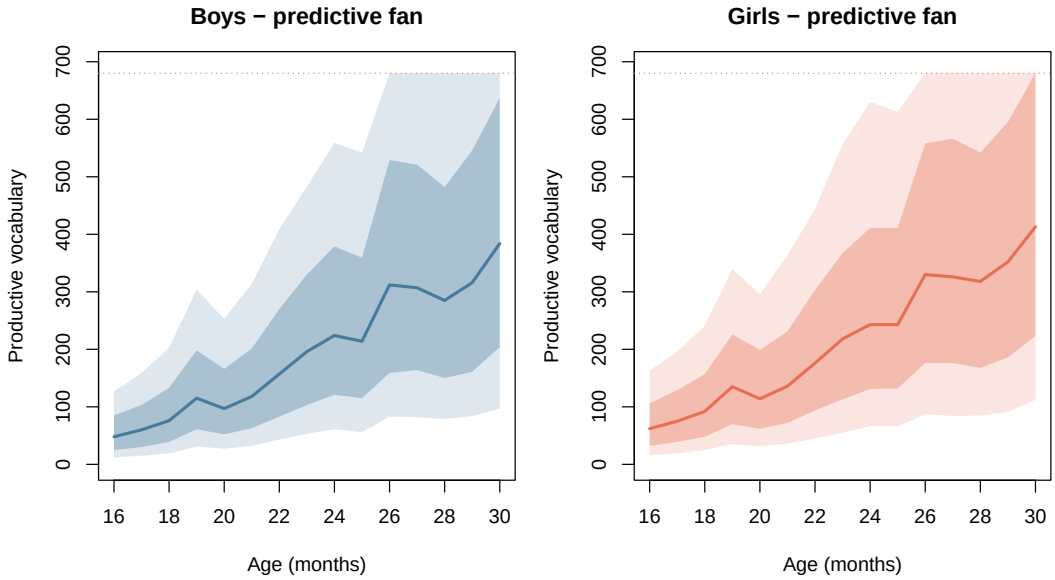


Figure 18: Posterior predictive "fan chart" for productive vocabulary across the full 16–30-month age range, separately for boys (left, blue) and girls (right, red). Shaded bands show the 10th–90th (light) and 25th–75th (dark) posterior predictive percentile ranges; the solid line is the predictive median. Both fans widen rapidly between 18 and 24 months (vocabulary explosion window) and are clipped at the 680-word ceiling at older ages, especially for girls whose higher central tendency makes ceiling contact earlier.

10.3 Three Developmental Phases

Phase 1 — Pre-explosion (18–20 months). Both distributions are strongly right-skewed, with most children producing fewer than 100 words and a long tail toward 200–250. The sex gap at the median is modest (17–20 words), while the 10th–90th band spans over 200 words, confirming that inter-individual variability dominates any group-level difference at this age. This is consistent with Bloom [1993]'s observation that the timing of vocabulary onset is the primary source of variability in early language development.

Phase 2 — Vocabulary explosion (22–24 months). This is the period of maximum inter-individual divergence: the 10th–90th band reaches over 500 words (59–565 for males at 24 months). The shape transitions from right-skewed to approximately uniform, reflecting the simultaneous presence of children who have not yet begun rapid vocabulary growth and children already near the CDI ceiling. The female advantage reaches approximately 24 words at the median (247 vs 223 at 24 months). An important clinical implication: a 24-month-old boy producing 100 words is not necessarily "delayed" — he is below the 25th percentile but the predictive distribution shows that a non-trivial fraction of typically developing children are in that range at this age.

Phase 3 — Ceiling approach (26–30 months). The 90th percentile is constrained at 680 for both sexes from age 26 onward (after truncation), reflecting the CDI ceiling effect. The median continues to rise (305 male, 335 female at 26 months; 385 male, 398 female at 30 months), but the upper tail is artificially compressed. Pre-truncation limits exceeded 680 (e.g., 969 for males at 30 months), indicating that the NegBin likelihood does not natively respect the CDI ceiling — a structural limitation noted in Section 11.

10.4 The Sex Gap in Clinical Perspective

The female−male median gap varies from 13 to 36 words across age groups. This gap is **statistically credible** (CI on γ excludes zero in NB3) but **clinically modest**: the 10th–90th bands of males and females overlap almost completely at every age. Table 14 is therefore best interpreted as a **sex-stratified normative reference** rather than evidence of a clinically actionable sex difference for individual children. The primary clinical utility is the 10th percentile threshold: a child at 24 months producing fewer than 59 words (male) or 64 words (female) falls in the lower tail of the predictive distribution and warrants further assessment.

Key Finding

Predictive conclusion. The Bayesian hierarchical NegBin model (NB3) yields sex-stratified posterior predictive distributions that can serve as normative

references for vocabulary screening. At 24 months — the most clinically relevant milestone — the model predicts a male median of 223 words (10th–90th: 59–565) and a female median of 247 words (64–624). The enormous width of these bands reflects genuine inter-individual variability, not model uncertainty: even a "perfect" model of average trajectories would face this irreducible aleatoric uncertainty, captured by the NegBin dispersion parameter $\hat{r} \approx 1.65$.

11 Discussion and Conclusions

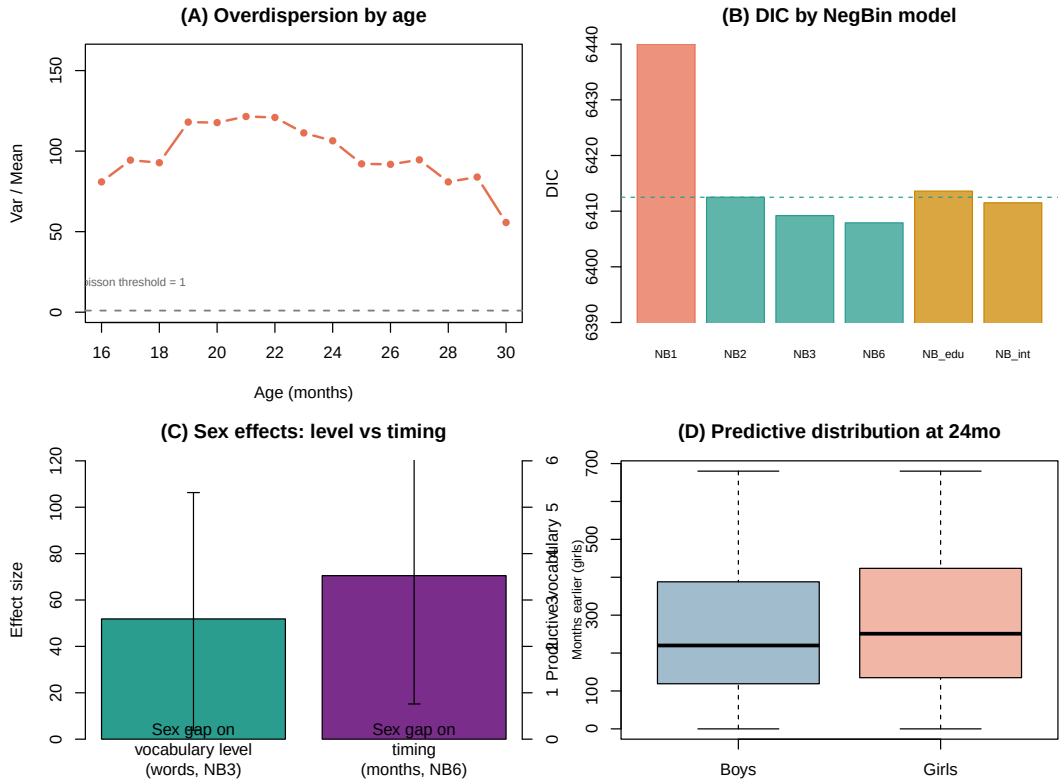


Figure 19: Executive summary of the four central findings. (A) Overdispersion ratio by age: all 15 points above the Poisson threshold of 1. (B) DIC by NegBin model with NB2 as reference. (C) Sex effects: 52-word level gap (NB3, green) and 3–4-month timing gap (NB6, purple), both with 95% credible intervals excluding zero. (D) Predictive distribution at 24 months: the bands of boys and girls overlap substantially despite the credible group-level difference.

11.1 Summary of Main Findings

This project has presented a fully Bayesian analysis of early vocabulary growth in 500 children aged 16–30 months (Marchman dataset, WordBank), using six Negative Binomial hierarchical models fitted via JAGS. The analysis was motivated by the recommendation to use count likelihoods for CDI data, confirmed by variance/mean

ratios of 55–122 across age groups.

Finding 1: The Negative Binomial likelihood is essential — and the Poisson is catastrophically inadequate. The Poisson baseline P0 — fitted with the identical mean structure as NB1 — achieves a DIC hundreds of points higher than every NegBin model despite perfect numerical convergence. The model is not wrong mathematically; it is wrong distributionally: the Poisson constraint $\text{Var}(y_i) = \mu_i$ is so severely violated (observed ratios 55–122) that it degrades predictive performance dramatically. The NegBin resolves this through its Poisson-Gamma mixture structure, adding $\mu_i^2/r \approx$ 60,000 words^2 of latent between-child heterogeneity at age 24 months that P0 forces to exactly zero. The estimated dispersion $\hat{r} \approx 1.4$–1.6 confirms massive overdispersion relative to Poisson across all models. The NegBin variance function $\text{Var}(y) = \mu + \mu^2/r$ successfully accommodates the extreme between-child variability documented in the descriptive statistics. Using a Normal or Poisson likelihood would constitute a fundamental misspecification of the data-generating process for CDI scores.

Finding 2: Hierarchical structure is the dominant modelling contribution. The transition from population-level NB1 to hierarchical NB2 yields $\Delta\text{DIC} = 87.3$ — by far the largest single improvement. Age-specific asymptotes α_j correctly resolve the ceiling overshoot of NB1 (which estimated $\hat{\alpha} = 644 > 680$), enabling biologically meaningful estimates: $\hat{\mu}_\alpha \approx 424$ words, $\hat{\beta} \approx 0.059$ (corresponding to 50% of the asymptote at approximately 20 months). The between-group heterogeneity $\hat{\sigma}_\alpha \approx 205$ words quantifies the enormous variability in asymptotic levels across monthly age groups documented by Fenson et al. [1994] and Frank et al. [2017].

Finding 3: Sex has a real but modest predictive contribution. Both NB3 ($\hat{\gamma} = 52$ words on α, CI excludes zero) and NB6 ($\hat{\delta}_\beta = 0.016$ on β, CI excludes zero) confirm the female advantage documented across languages [Fenson et al., 1994, Eriksson et al., 2012]. NB6 provides the more mechanistically informative result: girls approach their asymptotic vocabulary approximately 4 months earlier than boys (half-life at ≈ 19 vs. ≈ 22 months), consistent with Bates et al. [1994]'s input-sensitivity account and Huttenlocher et al. [1991]'s maturation account. However, DIC improvements of 3–5 points are below the threshold for strong evidence (Ntzoufaris, 2010), reflecting the dominance of within-age variability over between-sex differences in this sample.

Finding 4: Education and the age×sex interaction are not supported in this sample. The null result for caregiver education (NB_{edu}, DIC worse than NB2) is likely attributable to the relative SES homogeneity of the Marchman sample. Hart & Risley [1995] documented their vocabulary gap in a sample specifically recruited to cover the full SES spectrum (welfare to professional families), with proportionally far greater SES variation than the predominantly middle-class Marchman cohort. The age×sex interaction (NB_{int}) is biologically plausible [Bornstein et al., 2004] but statistically underpowered: with ∼33 children per monthly age group in the 500-child subsample,

the individual γ_j estimates have CIs spanning over 200 words.

Finding 5: Bayesian inference provides substantial advantages over frequentist alternatives. The Bayesian p-value of $p_B = 0.564$ confirms adequate model calibration. The comparison with per-group MLE demonstrates the Stein-James shrinkage effect [Efron & Morris, 1977]: Bayesian hierarchical estimates for data-poor age groups (16–22 months) are 70–150 words above MLE, reflecting principled regularisation toward the population mean. At data-rich age groups, the two approaches converge — the empirical confirmation of the Bernstein-von Mises theorem [**?**].

11.2 Limitations

- **Cross-sectional data.** Each child is observed at exactly one time point. This prevents estimation of individual growth trajectories and necessitates age-group-level (rather than child-level) hierarchical modelling. A longitudinal design would enable true random-effects growth curve modelling [Pinheiro & Bates, 2000], where α_i and β_i would be estimated for each child using their repeated measurements.

- **CDI ceiling.** The NegBin likelihood does not explicitly model the hard upper bound of 680 words. At ages 28–30 months, a substantial fraction of children approach or reach this ceiling, causing left-skewness that the NegBin can accommodate through the mean structure but not through a formal truncation mechanism. A Tobit or censored regression framework would handle this more formally.

- **Sample homogeneity.** The Marchman dataset provides good internal validity but limited SES and ethnic diversity. This reduces the statistical power to detect education (Hart & Risley, 1995) and birth-order (Hoff, 2003) effects. An exploratory multi-dataset analysis with a dataset-level random effect yielded $\Delta\text{DIC} = 132.7$ vs. the Marchman-only NB2, confirming that substantial between-dataset heterogeneity exists and that a larger, more heterogeneous sample would provide more power for covariate detection.

- **Birth order missing data.** The 43% missing rate for birth_order prevented its inclusion as a formal model covariate despite documented effects [Hoff, 2003]. Bayesian multiple imputation within JAGS — treating missing values as latent variables with a Bernoulli prior — is technically feasible and was explored in the analyses but not included in the main model comparison due to the non-comparability of DIC across models with different missing-data structures.

11.3 Future Directions

- **Multi-dataset hierarchical model** with a dataset-level random effect $\delta_d \sim \mathcal{N}(0, \sigma_d^2)$, enabling use of all 9,093 WordBank observations while formally accounting

for between-dataset heterogeneity. This would also restore power to detect SES (Hart & Risley, 1995) and bilingual (Pearson et al., 1993) effects.

- **Longitudinal modelling** with child-level random effects on α_i and β_i, requiring repeated CDI administrations to the same children.
- **Censored Negative Binomial** or Beta-Binomial likelihood to address the CDI ceiling at 680 words more formally.
- **Bayesian birth-order imputation** as part of the main model to recover this clinically relevant predictor [Hoff, 2003], using $\text{firstborn}_i \sim \text{Bernoulli}(\pi)$ with $\pi \sim \text{Beta}(1, 1)$.

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